Domain-Specific Image Generation with LoRA

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Part 2 findings - DRAFT: remaining submission items are listed at the end

Abstract

This experiment compares Stable Diffusion 1.5 with two attention-only LoRA adapters, ranks 4 and 8. Both use the same cleaned dataset, split and update budget. Rank 4 gives the lowest KID, while the base model keeps the highest overall CLIP alignment. The remaining human study and submission steps are clearly identified.

I. Data and experimental setup

I use Naruto BLIP Captions, an anime-frame dataset with automatically generated captions. Exact duplicate pixels, corrupt files and empty captions are removed before the seed-266 90/10 split. Images are converted to RGB and resized to 512 x 512. Test images are reserved for evaluation.

Verified preparation: {"corrupt": 0, "duplicate": 0, "empty_caption": 0}; 1098 train and 123 test images. The saved output reports no removed corrupt, duplicate or empty-caption samples.

Two important limits are narrow anime/character coverage and noisy captions. In the prepared data, some captions repeat words or describe clothing vaguely. These issues can weaken prompt following and limit generalization to photographs or other domains.

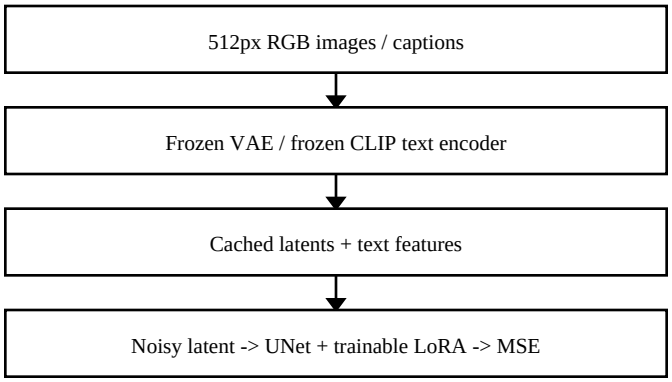


Fig. 1. Training architecture. Frozen components are run before the UNet to reduce peak memory.

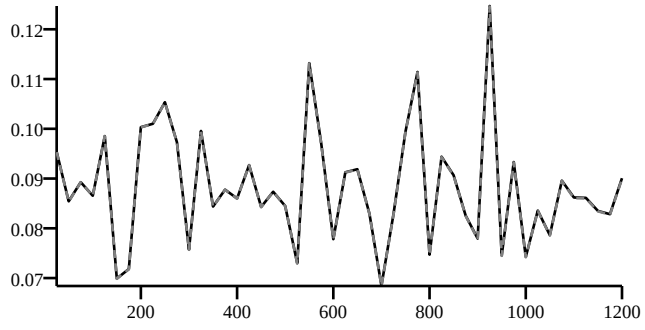
II. Controlled training comparison

Both ranks use 1200 AdamW updates at learning rate 0.0001, microbatch 1 and accumulation 4. Only attention LoRA weights update. Base weights stay frozen, gradient checkpointing is enabled, and fp16 is used for the frozen UNet. The lab requires a working single-16-GB-GPU path. A successful 24-GB RTX 4090 setup alone does not demonstrate that memory target.

Run	*Rank*	*Trainable*	*LR*	*Updates*	*Final MSE*
run1	*4*	*797184*	*0.0001*	*1200*	*0.0394*
run2	*8*	*1594368*	*0.0001*	*1200*	*0.0393*

Run	*Batch/acc.*	*Peak MiB*	*Seconds*	*GPU-h*
run1	*1/4*	*1949.9*	*764.5*	*0.2124*
run2	*1/4*	*1960.9*	*773.0*	*0.2147*

Table I. Measured optimizer-loop cost, including checkpoint writes but excluding shared VAE/text caching and evaluation.



Optimizer updates; solid: rank 4, dashed: rank 8

Training noise MSE, averaged in blocks of 25 updates. This is not a validation quality score.

III. Generation and evaluation

Twenty prompts are frozen before training: ten held-out captions, five out-of-domain requests and five compositional requests. Base, rank 4 and rank 8 each generate seeds 0 and 1 using the same sampler, step count and guidance settings: 120 images in total. The safety checker remains enabled. Eight grids compare the models side by side.

Sampling protocol: {"scheduler": "DDIMScheduler", "steps": 20, "guidance_scale": 7.5, "eta": 0.0, "height": 512, "width": 512, "negative_prompt": ""}

KID compares the twenty in-domain generations per model with the test images; lower is better, but the estimate is noisy at this size and may be negative. CLIP score measures image-prompt alignment. LPIPS measures diversity across the two seeds for each prompt. None of these replaces checking whether characters, counts and relationships are actually correct.

Model	*KID*	*CLIP*	*LPIPS*
base	*0.1274*	*31.24*	*0.6529*
run1	*0.0681*	*29.54*	*0.7547*
run2	*0.0941*	*29.49*	*0.6462*

Table II. Actual saved automatic metrics. KID uses in-domain images only; CLIP and LPIPS cover all twenty prompts.

Rank 4 is closer to the held-out domain under KID than either the base or rank 8. However, the base has the strongest average CLIP alignment. Fine-tuning therefore improves one goal without improving every goal. Increasing the rank does not help this experiment enough to justify its larger adapter.

Run1 has the largest LPIPS distance, but this needs caution: one black filtered image paired with a normal image can also raise perceptual distance. That score alone is not proof of more useful diversity. KID uses only twenty generated in-domain images, so its subset variation is not a confidence interval for repeated training runs.

Model	*Filtered / 40*
base	*0 / 40*
run1	*6 / 40*
run2	*7 / 40*

Table III. Safety-filtered images remain in the evaluation. They are not silently replaced with more favorable samples.

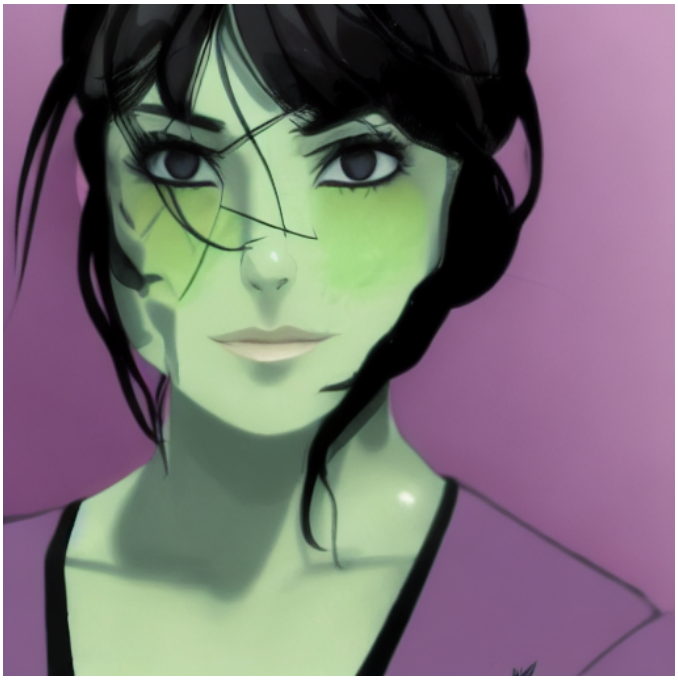
IV. Human evaluation and model choice

The program creates 25 blinded comparisons with randomized left/right placement. Two team members and at least one person outside the team must independently judge domain style and prompt match. The reported win rate is final-run wins divided by all judgments, with ties reported separately. Human judgments are not generated by an LLM.

The independent three-rater comparison is not complete, so no human win rate is reported. The visual observations below are AI-assisted image review, not a substitute for those ratings.

I would choose run1, the rank-4 adapter, for the domain-style model. It has the lowest KID of the three models and a slightly higher CLIP score than run2, while using half as many trainable adapter parameters. The visible portrait examples also show the intended move from photographic faces toward an anime style. This is a trade-off, not a win on every measure: the base model has the highest overall CLIP score. Run1's larger LPIPS value is not enough to claim better useful diversity, because safety-filtered black outputs can affect that measurement. The choice is based on automatic metrics and an AI-assisted inspection of the saved grids; the independent human comparison is still pending.

V. Failure analysis



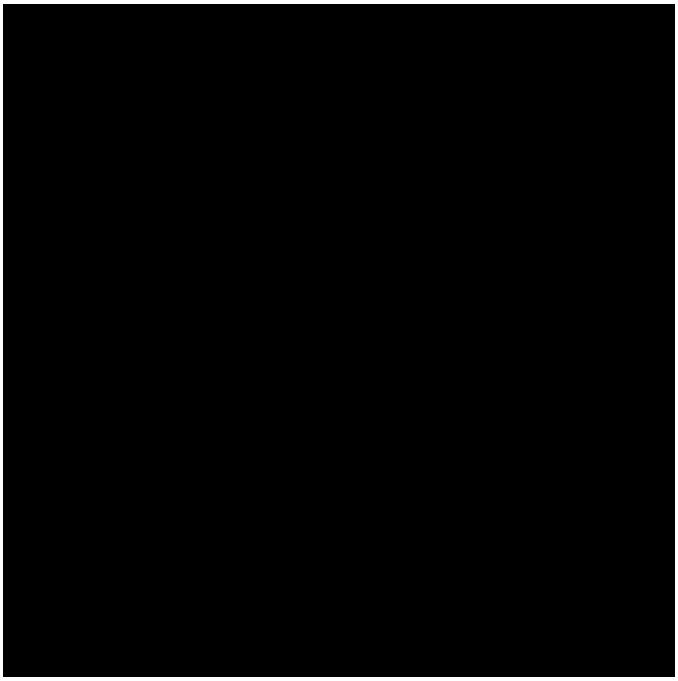
Failure example 1: run1_02_0.png

The prompt asks for a woman with green eyes and black hair. In the rank-4 panel the hair is dark, but a strong green tint covers much of the face while the eyes look dark or gray rather than clearly green. The image adopts an anime style without reliably keeping the requested color attached to the eyes. This suggests a prompt-attribute binding problem. Fine-tuning may strengthen familiar colors and portrait patterns without improving which object receives each attribute. That explanation is a hypothesis from the visible example, not a proven mechanism. More varied, better-captioned training examples and a separate attribute-following evaluation would help test it.



Failure example 2: run1_04_0.png

The prompt asks for a group in a field with one person holding a sword. In the rank-4 panel, a foreground character and a background character both appear to hold blades. The field and group idea are present, but the 'one person' relationship is not preserved. The base comparison also has several visible weapons, so this is not a failure unique to LoRA. The model appears to capture the general scene more easily than the exact count and ownership of the swords. A single CLIP similarity score can overlook this kind of compositional error. A useful follow-up would score the number of sword-holders explicitly and improve captions that describe counts and relationships.



Failure example 3: run1_05_0.png

For the prompt about a man with long hair and a brown jacket, both fine-tuned panels are completely black. The saved generation logs report safety filtering, and the aggregate counts are six filtered outputs for run1 and seven for run2. These panels provide no usable visual answer to the prompt. The enabled safety checker replaces flagged generations with black images. The underlying unfiltered images are not available, so the black panels alone do not establish whether those flags were correct or false positives. Keep the filter enabled, report the missing usable outputs, and do not silently regenerate until a preferred result appears.

VI. Safety, licensing and memorization

The Naruto dataset card does not include an explicit license field. Instructor approval of a dataset option does not itself grant copyright or redistribution rights. Anime frames may contain copyrighted characters and designs, and generated images may reproduce similar features. Before uploading adapters or samples, confirm the intended educational sharing is permitted. The base SD 1.5 model has its own CreativeML Open RAIL-M conditions; they do not resolve source-image rights.

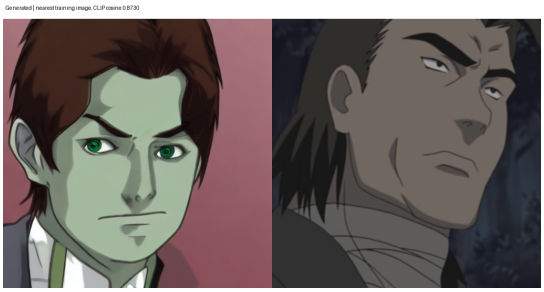
The course describes Naruto and Pokemon characters as copyrighted and limits this model to non-commercial research. This report follows that scope and does not claim that permission grants ownership of the source images.

The diffusion pipeline keeps its NSFW checker and records its flag for every generation. A filtered output remains counted rather than being silently replaced. The checker can make mistakes, so manually inspect outputs before sharing. Avoid unsafe, deceptive or real-person-targeted prompts. Use private artifact repositories by default and share access with the graders as required. Private sharing reduces exposure but does not create a license.

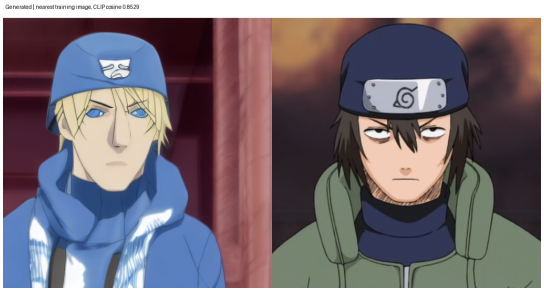
The evaluation code embeds ten seed-0 in-domain generations from each adapter with CLIP, compares them with all training images, and saves the three highest-similarity pairs. High cosine similarity is a warning for inspection, not proof that an image was copied. Low similarity is not proof that memorization is absent. The check is limited by ten samples, a single embedding model and a small training set. A stronger check would use more prompts, perceptual hashes and direct visual review.

Do not publish rater identities or authentication tokens. The human-rating form uses anonymous IDs, and the token is entered using a hidden prompt. Both team members still need to inspect the code, run the experiment and explain the modeling and safety decisions during the demo.

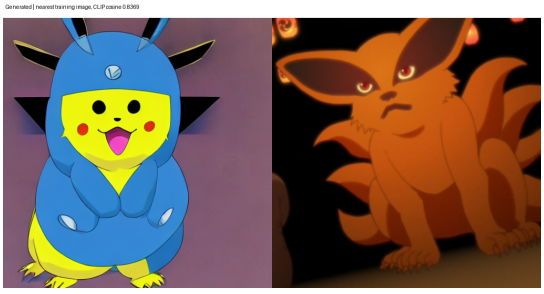
For the selected rank-4 model, the closest three generated/training pairs have CLIP cosine similarities 0.8730, 0.8529, 0.8369. I can see shared anime-style features, but the displayed pairs differ in their hair, poses or characters and are not exact visible copies. This small inspection does not establish that memorization is absent.



Closest training pair 1 for run1; generated image left, training image right.

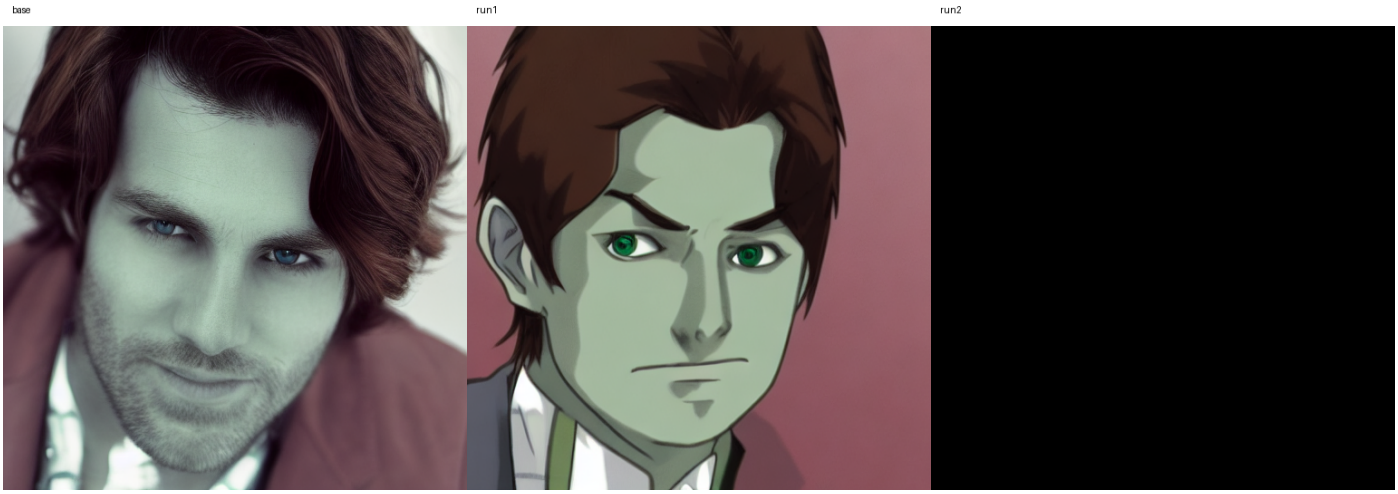


Closest training pair 2 for run1; generated image left, training image right.

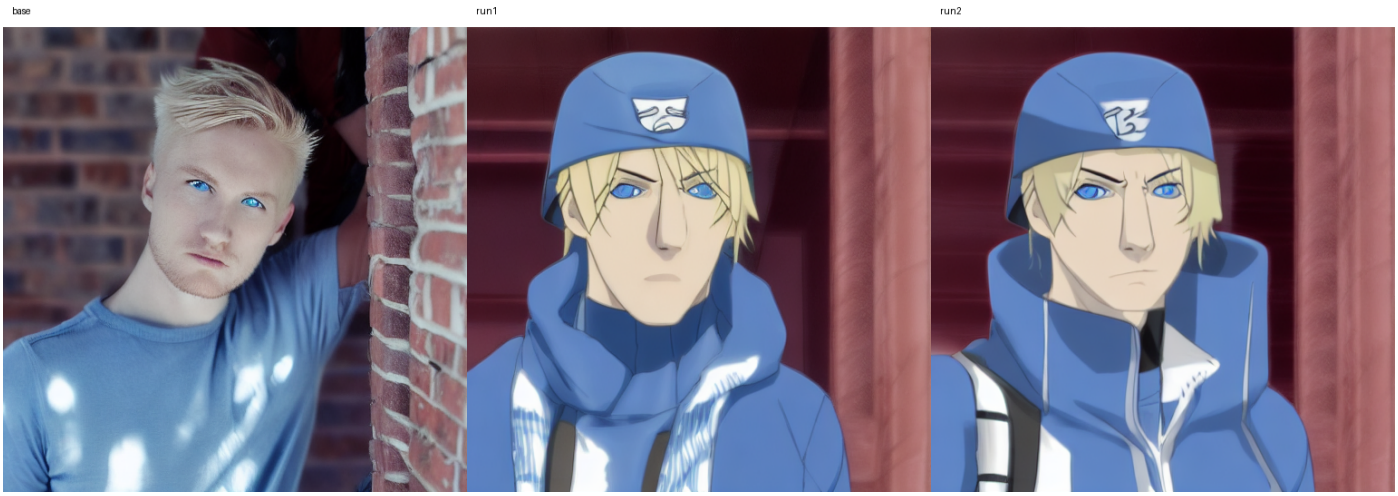


Closest training pair 3 for run1; generated image left, training image right.

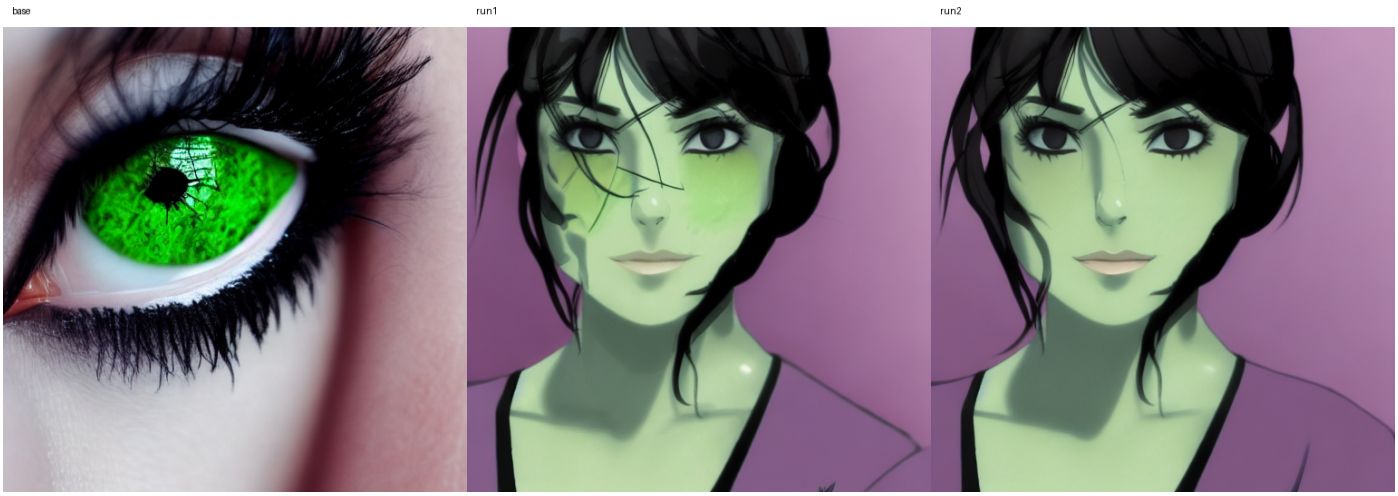
Appendix: controlled visual comparisons



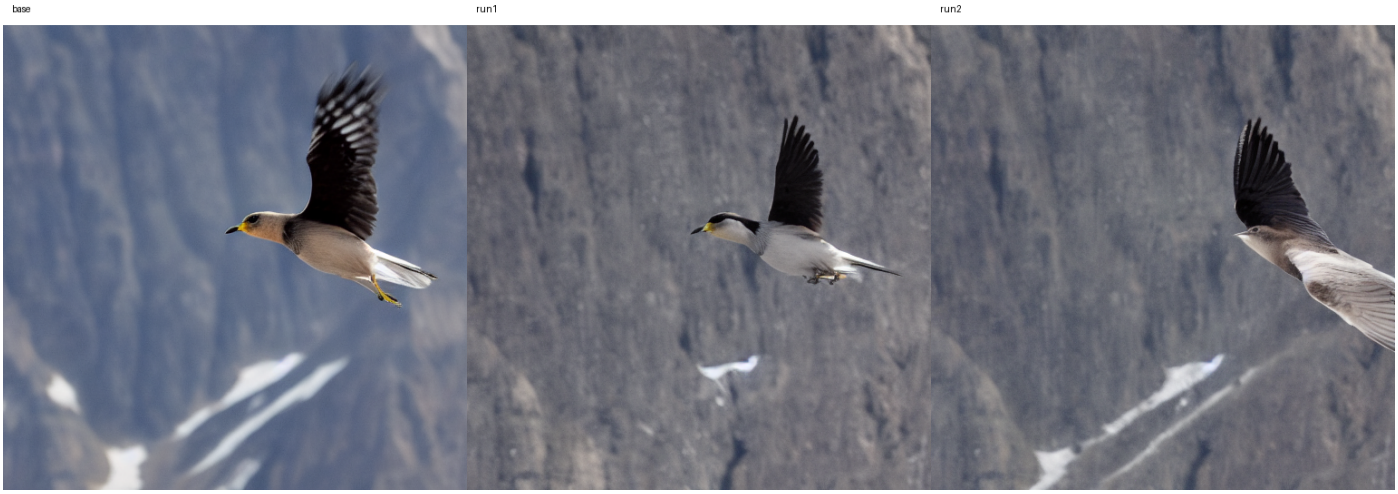
Prompt 1, seed 0: base, rank 4 and rank 8, left to right.



Prompt 2, seed 0: base, rank 4 and rank 8, left to right.



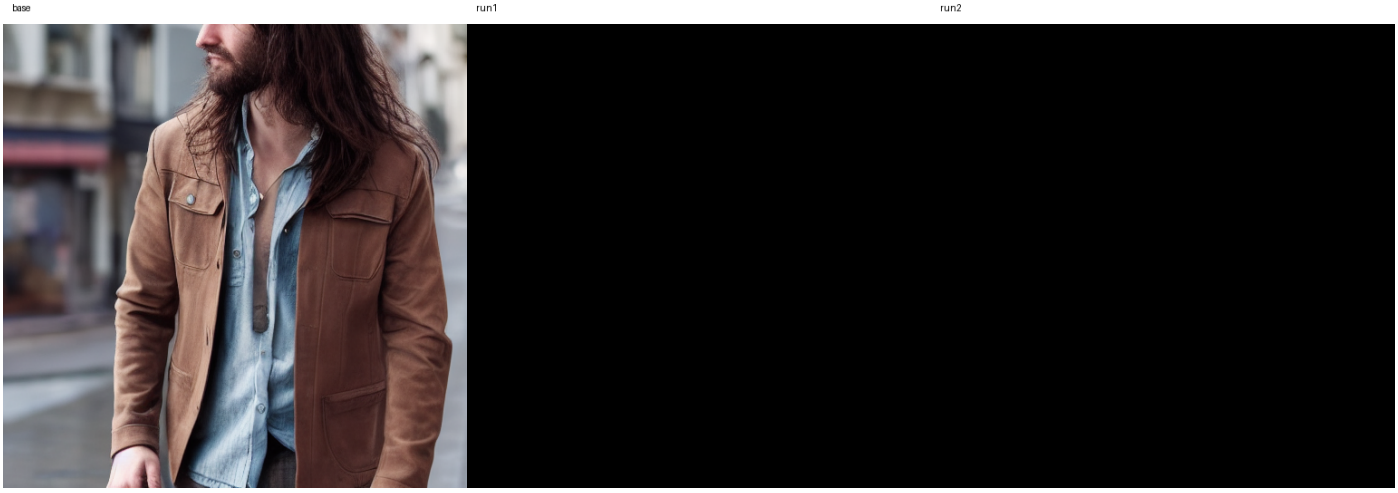
Prompt 3, seed 0: base, rank 4 and rank 8, left to right.



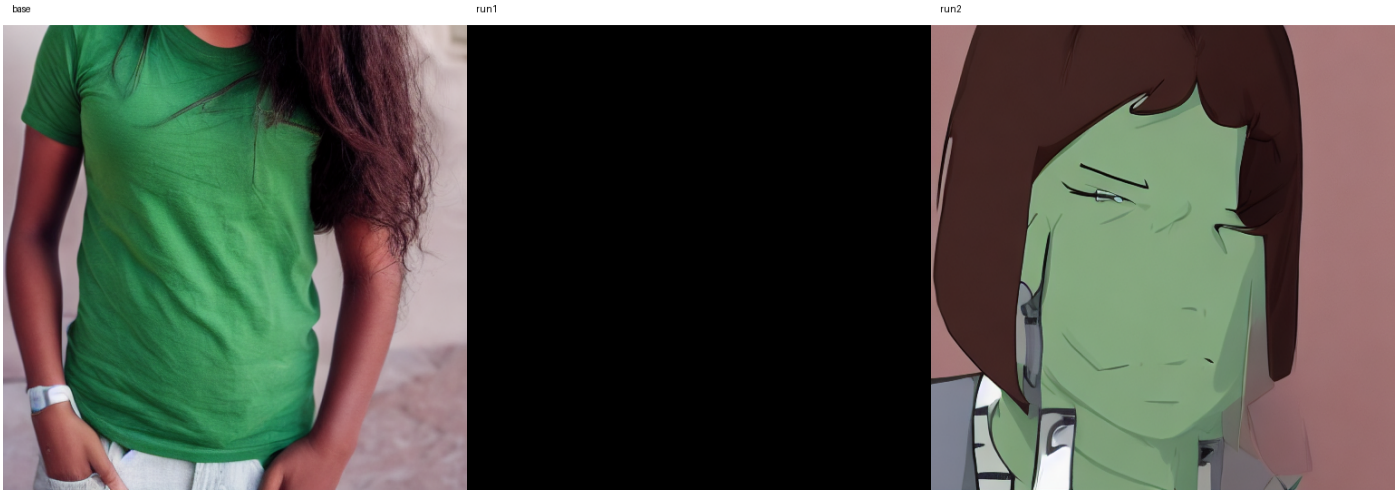
Prompt 4, seed 0: base, rank 4 and rank 8, left to right.



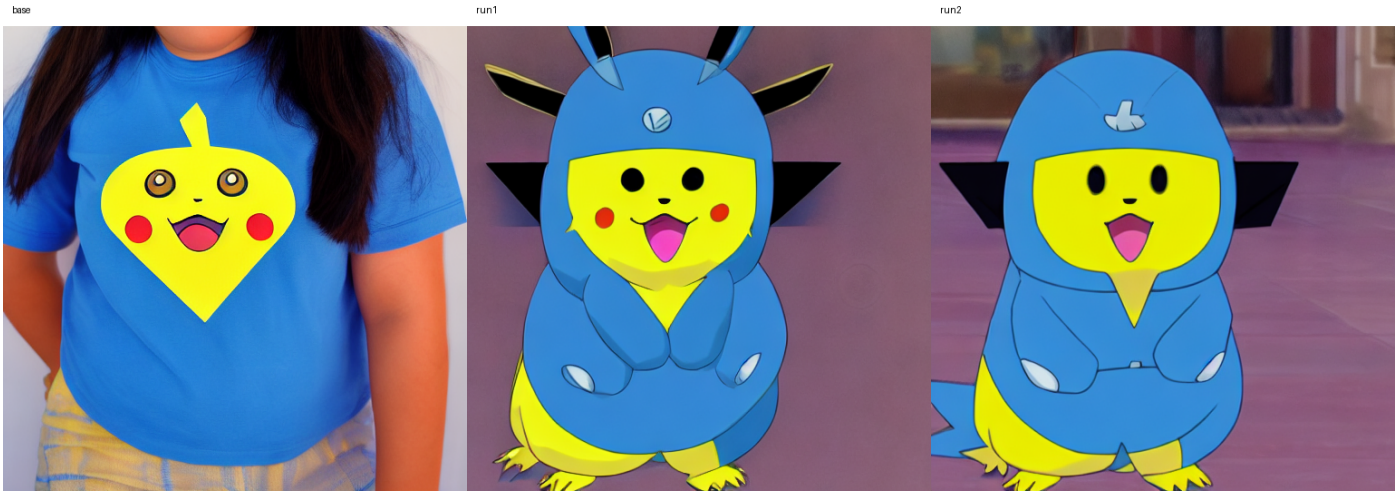
Prompt 5, seed 0: base, rank 4 and rank 8, left to right.



Prompt 6, seed 0: base, rank 4 and rank 8, left to right.



Prompt 7, seed 0: base, rank 4 and rank 8, left to right.



Prompt 8, seed 0: base, rank 4 and rank 8, left to right.

Submission status

This is a Part 2 section draft, not the required final combined report for all three parts. The final submission also needs team contributions, AI disclosure, discussion recordings, the tag commit links, and the Kaggle link. Pending items are listed below so incomplete work is not mistaken for a finished submission.

Pending: 25 blind pairs 3 real raters

Pending: artifact access for graders verified

Pending: team details recorded

Pending: license and redistribution reviewed

run1: https://huggingface.co/denishatank/data266-lab2-p2-run1

run2: https://huggingface.co/denishatank/data266-lab2-p2-run2

images: https://huggingface.co/datasets/denishatank/data266-lab2-p2-images

References

[1] DATA 266 Lab 2 requirements, pp. 8-9 and 11-12, provided by the course.

[2] Rombach et al. High-Resolution Image Synthesis with Latent Diffusion Models, 2022. arxiv.org/abs/2112.10752.

[3] Hu et al. LoRA: Low-Rank Adaptation of Large Language Models, 2021. arxiv.org/abs/2106.09685.

[4] Carlini et al. Extracting Training Data from Diffusion Models, 2023. arxiv.org/abs/2301.13188.